

**Linear probes: open-weight models encode eval-vs-deploy context early in the residual stream (35 models, SAD)**

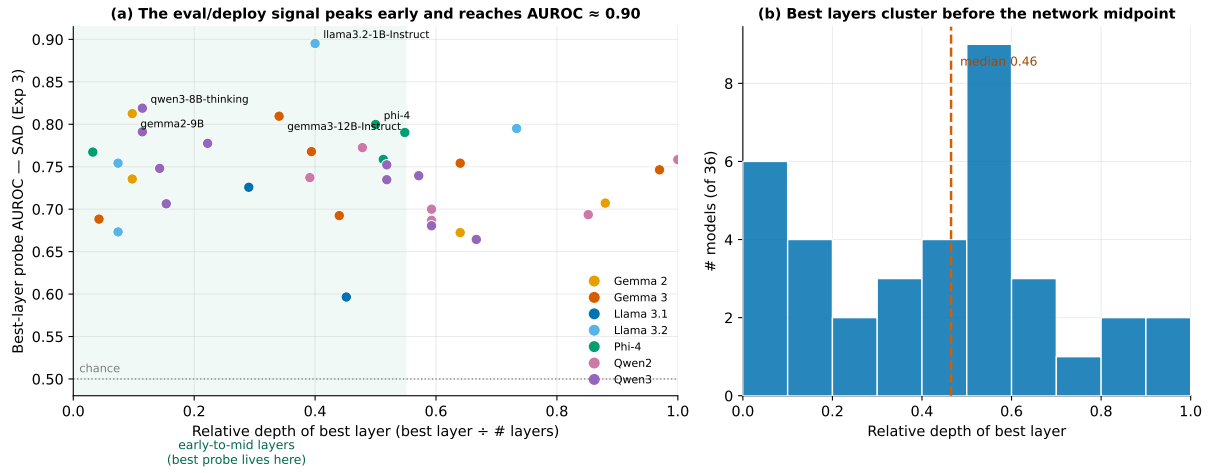


Figure 14: **Eval/deploy context is encoded early in the residual stream.** (a) Best-layer linear-probe AUROC on the SAD stages-oversight dataset (Experiment 3) vs. relative depth of the best layer (best layer  $\div$  total layers), for 35 open-weight models spanning seven families. The shaded region highlights early-to-mid layers (relative depth  $\lesssim 0.55$ ). Peak probe AUROC reaches 0.8601 (Llama-3.2-1B-Instruct, layer 6). (b) Histogram of relative best-layer depth across all models; the median is 0.46, i.e., before the network midpoint. Points are colored by model family (Okabe–Ito palette).

Figure 14 characterizes *where* the eval-awareness signal lives in model depth. Across 35 models, the strongest linear probe for distinguishing evaluation from deployment contexts peaks in early-to-mid layers rather than at the final residual-stream positions: the median relative depth of the best layer is 0.46, and very few models achieve peak probe performance beyond the network midpoint. Despite this early localization, probe quality can be high-up to AUROC 0.8601 on SAD-indicating that eval-vs.-deploy discrimination is a salient, linearly separable feature of internal representations long before it is fully expressed in behavioral outputs. This depth profile is consistent with eval/deploy cues being processed early and subsequently transformed into task-specific features in deeper layers, and it complements the latent-vs.-behavioral gap shown in our main probing figure.

| Model                  | AUC   | Acc.  |
|------------------------|-------|-------|
| Qwen3-14B non-thinking | 0.714 | 0.660 |
| Qwen3-14B thinking     | 0.701 | 0.658 |
| Gemma2-27B-Instruct    | 0.693 | 0.612 |
| Qwen3-8B thinking      | 0.682 | 0.652 |
| Gemma2-9B-Instruct     | 0.676 | 0.634 |
| Qwen3-8B non-thinking  | 0.675 | 0.664 |
| Qwen3-4B non-thinking  | 0.667 | 0.656 |
| Llama3.1-8B-Instruct   | 0.644 | 0.606 |
| Gemma3-12B-Instruct    | 0.624 | 0.618 |
| Human annotators       | 0.819 | 0.752 |
| GPT-4.1                | 0.760 | 0.747 |
| Claude 3.5 Sonnet      | 0.789 | 0.774 |
| Random                 | 0.500 | 0.500 |

Table 15: Representative Needham chat-subset detection results.

| Model                 | AUC   | Acc.@0.5 |
|-----------------------|-------|----------|
| Gemma3-1B-Instruct    | 0.622 | 0.417    |
| Qwen3-4B-thinking     | 0.554 | 0.512    |
| Qwen3-4B-non-thinking | 0.541 | 0.509    |
| Qwen2-0.5B            | 0.530 | 0.573    |
| Llama3.2-3B           | 0.524 | 0.519    |
| Llama3.2-1B           | 0.514 | 0.544    |
| Gemma2-9B-Instruct    | 0.508 | 0.430    |
| Qwen2-7B              | 0.498 | 0.485    |
| Qwen3-8B-non-thinking | 0.459 | 0.405    |
| Phi-4-mini-instruct   | 0.329 | 0.348    |

Table 16: Representative  $2 \times 2$  design results from 32 validated model runs.

## H Supplementary Result Tables

These tables give the full numbers for results summarized inline in Section 5: Needham detection (Tab. 15), the  $2 \times 2$  format/intent design (Tab. 16), SAD stages-oversight rewrite accuracy (Tab. 17), cross-format probe survival (Tab. 18), multi-layer steering (Tab. 19), and the cross-experiment paradigm correlations (Tab. 20).

| Model                | Tier 0 | Tier 2 | Tier 2-Sub1 |
|----------------------|--------|--------|-------------|
| Gemma3-4B-Instruct   | -0.073 | -0.143 | -0.298      |
| Gemma2-9B-Instruct   | -0.065 | -0.148 | -0.238      |
| Llama3.1-8B-Instruct | -0.163 | -0.155 | -0.170      |
| Gemma3-12B-Instruct  | -0.082 | -0.023 | -0.115      |
| Gemma3-1B-Instruct   | +0.158 | -0.003 | -0.008      |
| Qwen3 variants       | 0.000  | 0.000  | 0.000       |

Table 17: SAD stages-oversight rewrite accuracy deltas relative to original.

Table 18: Cross-format probe survival (gemma2-9B-Instruct). A logistic-regression probe trained only on original prompts keeps AUROC  $\geq 0.98$  on the SAD rewrites that drive the *same* model’s behavioral accuracy below chance - the attack destroys the answer, not the representation. A diff-of-means probe stays near chance, ruling out item leakage.

| SAD variant | Behavioral acc. | Probe AUROC |
|-------------|-----------------|-------------|
| original    | 0.575           | 0.996       |
| tier 0      | 0.510           | 0.983       |
| tier 2      | 0.427           | 0.991       |
| tier 2-Sub1 | 0.338           | 0.991       |

Table 19: Multi-layer steering of Llama-3.1-8B-Instruct (layers 4 · 9 · 15) along the contrastive eval/deploy direction is causal across three independent downstream tasks at  $\alpha = +0.8$ ; matched random-vector controls reproduce none of the effects. \*  $p < 0.001$  (McNemar / permutation).

| Task              | Effect                   | learned | control |
|-------------------|--------------------------|---------|---------|
| Needham detection | $\Delta AUROC$           | -0.123* | +0.009  |
| HarmBench safety  | $\Delta ASR$ ( $n=200$ ) | +0.075* | -0.025  |
| SAD oversight     | discrimination           | +0.945  | -0.235  |

Table 20: Spearman correlations between paradigm metrics (per-model,  $n=32-36$ ). Only the awareness-safety pair is significant; it is *size-independent* (partial  $\rho = -0.77$  controlling for log-parameters;  $\rho = -0.74$  within instruction-tuned models alone). The remaining pairs are near-orthogonal.

| Paradigm A  | Paradigm B        | $\rho$ | $p$     |
|-------------|-------------------|--------|---------|
| Needham AUC | HarmBench framing | -0.79  | < 0.001 |
| Needham AUC | 2 × 2 AUC         | -0.18  | 0.32    |
| Needham AUC | probe AUROC       | -0.11  | 0.54    |
| 2 × 2 AUC   | probe AUROC       | +0.02  | 0.90    |

Table 21: Needham eval-awareness detection: steered vs. control vs. baseline (Llama-3.1-8B-Instruct, layers 4-9-15,  $n=500$ ). Activation steering reduces the model’s ability to distinguish eval from deployment contexts. %Yes(eval) and %Yes(real) denote the fraction of prompts classified as evaluations under eval and deployment contexts respectively. Negative  $\alpha$  suppresses both rates toward zero; positive  $\alpha$  inflates both toward one, compressing the eval/real gap in either direction. Matched random-vector controls produce weaker and less consistent shifts.

| Condition | $\alpha$ | AUC   | % Yes (eval) | % Yes (real) | disc (pp) | $\Delta$ disc |
|-----------|----------|-------|--------------|--------------|-----------|---------------|
| baseline  | 0.0      | 0.676 | 78.7         | 67.1         | +11.5     | —             |
| steered   | -0.8     | 0.601 | 51.8         | 46.3         | +5.5      | -6.0***       |
| control   | -0.8     | 0.608 | 84.9         | 78.6         | +6.3      | -5.3***       |
| steered   | +0.8     | 0.552 | 85.2         | 82.8         | +2.4      | -9.1***       |
| control   | +0.8     | 0.685 | 71.0         | 57.5         | +13.5     | +2.0          |

\*\*\*  $p < 0.001$  (permutation test on disc). disc = %Yes(eval) - %Yes(real).